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Executive function profiles identify stable schizophrenia subtypes with distinct symptoms, biological markers and treatment responses: a cross-time clustering and validation study

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Abstract

Background Executive function (EF) is a heterogeneous neuropsychological construct, and impairments in EF dimensions represent a core aspect of psychopathology in schizophrenia that varies across individual patients. Currently, how this inter-individual variability characterizes schizophrenia subgroups, along with their distinctions in clinical characteristics and prognostic outcomes, remains unclear.

Methods Three EF dimensions (inhibitory control, working memory, cognitive flexibility) were assessed in the main sample ($N = 329$), its follow-up subset, and an independently "recurring local validation" patient sample ($N = 114$). Fuzzy clustering was applied to baseline EF assessments to discover and validate the core subtypes after excluding cluster-ambiguous cases in the main and independent samples, respectively. Subtype-based classification trained on the main sample was then tested in the independent sample. Importantly, the stability of these subtypes and their remission statuses, along with associated longitudinal changes in clinical and biological factors, were evaluated, and baseline subtype statuses were also used to predict outcomes.

Results Two longitudinally stable, independently validated core EF subtypes were identified, with Subtype I exhibiting selective deficits limited to response inhibition and cognitive flexibility ("EF-Focal") and Subtype II showing broad EF impairments across all dimensions ("EF-Global"). Compared to the "EF-Global" subtype, patients within the "EF-Focal" subtype presented significantly milder positive, affective, and cognitive symptoms, worse working memory updating, lower inflammatory markers, and better metabolic profiles at baseline. Models trained on baseline EF assessments from the main sample accurately classified patient subtypes in the independent sample (area under the curve [AUC] = 0.79); these baseline subtypes moreover classified patients' remission statuses independently of overall severity at intake (AUC = 0.78). Remitted patients were observed to have significantly greater reductions in negative

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and cognitive symptoms, improved working memory maintenance, lower peripheral inflammatory levels, and superior metabolic functions over time.

Conclusions EF subtyping was associated with symptomatic, biochemical, and prognostic variations in individuals with schizophrenia, which could help to stratify patients within this disorder for targeted treatments.

Keywords Executive dysfunction, Machine learning clustering, Prognostic prediction, Peripheral immune-metabolic, Schizophrenia, Subtyping

Background

Cognitive impairment is prevalent in patients with schizophrenia and is a core aspect of psychopathology [1]. This deficit emerges in the prodromal phase and persists throughout the illness but is difficult to treat using current antipsychotics [2]. Symptoms behaviorally measurable by clinical scales, such as the Positive and Negative Syndrome Scale (PANSS), are thought to arise from the impairments in functions related to memory, attention, and executive tasks [3, 4]. Executive functioning (EF) represents a series of higher-order cognitive processes, which involves impulse control and behavior orchestration [5]. Deficits in EF thus could hinder goal-directed activity and contribute to aggression, violence, and poor compliance with medication, leading to worse clinical outcomes [6]. While EFs are reported to be consistently affected in patients with schizophrenia, these effects have high inter-individual variability (ranging from mildly to severely impaired) [7]. This heterogeneity may allow for differentiating patient subgroups by EF severity to capture different clinical outcomes of schizophrenia. However, to date, a prediction of patient prognostic status in this disorder from baseline data remains challenging.

Cognitive dysfunction is widely considered a core psychopathology of schizophrenia that spans the entire disease trajectory (emerges during prodromal phases, persists throughout illness, and has implications for prognosis) [2, 8–10]. It often precedes the onset of positive and negative symptoms—a key differentiation from other major psychiatric disorders such as bipolar disorder and depression, in which cognitive deficits do not present before symptom expression [11]—and can influence their emergence and severity [12, 13]. Indeed, cognitive deficits including executive dysfunction are frequently detectable as early as childhood [14, 15], are predictive of long-term prognosis [16, 17], and "disorganized thinking" (a cognitive symptom manifestation linked to executive functions) remains a key diagnostic criterion [18]. This evidence highlights that cognitive function shares a fundamental, and potentially causal, link with the disease process itself. Crucially, cognitive functions are unevenly affected in the schizophrenia population, and such heterogeneity holds promise for evaluating and indicating clinical outcomes and real-world functioning of this disorder [19, 20]. Consequently, multiple attempts have been made to investigate cognitive subtypes in schizophrenia using clustering analysis; these attempts used features from comprehensive neuropsychological assessment batteries, such as the Measurement and Treatment Research to Improve Cognition in Schizophrenia (MATRICS) Consensus Cognitive Battery and the Brief Assessment of Cognition in Schizophrenia (BACS) [21]. Besides the commonly identified near-normal (intact except for EF) and globally (or multidomain) impaired cognitive subgroups, there are intermediate-decline subgroups differentiated by the expression patterns on EF dysfunction [22]. These cognitive subgroups are shown to remain stable over time, except for the multidomain-deteriorated subgroup, with nearly half of affected patients transitioning into the executive-deteriorated subgroup [23]. Furthermore, the deficits of EF in schizophrenia involve multiple aspects, including heightened challenges in inhibiting automatic responses and switching to new responses, diminished cognitive flexibility, and disruptions in maintaining and updating goal-related or rule-based information in working memory [24, 25]. However, previous studies did not consider these dimensions in their assessments of EF when deriving cognitive subgroups (cf. Additional file 1: Table S1) [23, 26–38]. Rather, they have anchored on a general, unitary perspective of EF (e.g., composite scores) [23, 27, 28, 37], as a domain of cognition for clustering, or their measured EF have been a reflection of single aspects of the multidimensional construct of EF [29–35] that requires a comprehensive assessment in essence. Importantly, EF deficits have a close link with the multidimensional symptoms in schizophrenia; specifically, patients with the positive and deficit/disorganized symptom subtypes present differentiated cognitive profiles as delineated by executive/attentional functions [32]. Also, EF deficits may contribute to the varying degrees of individual positive, negative, and disorganized symptoms, reflecting their heterogeneous role in psychopathology [39–41]. Although some dimensions of EF can be more severely affected than others in patients with schizophrenia [42], their characterized EF subtypes remain unclear, with added concerns related to distinctions in symptomatology, longitudinal stability, and the implications for prognostic outcome and remission status.

In the field of psychiatry, prognostic prediction is crucial for early assessment and intervention but remains a significant challenge given prior investigations of entire patient groups employing baseline data mostly have approached chance-level accuracy [43, 44]. In turn, subtype-based prediction may be more promising in facilitating timely treatment allocation of patients through stratified medicine [45]. Importantly, cognitive subgroups correlate with clinical outcomes along a spectrum—patients with the greatest deterioration display the highest severity and longest illness duration, whereas those least affected have milder outcomes and shorter duration [23, 27, 37]. Furthermore, EF alone has been frequently suggested to relate to a greater frequency of sustainable remission in patients with schizophrenia [46]. This relationship is also evident for EF dimensional assessments given that performance in EF dimensions differs between remitted and non-remitted patient groups, with previous studies highlighting working memory and inhibitory function as markers for long-term remission status [47, 48]. As a consequence, identifying EF subtypes based on baseline multi-dimensional assessments may hold promise for accurately predicting the prognostic status of patients with schizophrenia. Also, it is unclear how patients in subtypes stratified by varying degrees of deficit in EF dimensions experience a change in EFs and symptoms as well as other biological factors over time to drive a specific remission outcome. In this context, blood immune and metabolic indicators can be highlighted as relevant. Individuals with schizophrenia often show an altered inflammatory state [49] and lipid metabolism [50]. These may underlie the pathological processes of this disorder [51] and contribute to symptom manifestations [52]. Interestingly, lipid metabolites and inflammatory markers like C-reactive protein (CRP) may precede psychosis onset, fluctuate with illness phases [53], and correlate with clinical outcomes such as changes in PANSS negative subscale scores [54]. Furthermore, there are subgroups in schizophrenia with varying levels of inflammation, and such distinction is linked to cognitive impairments, including in working memory and response inhibition [55]. Moreover, antipsychotics influence peripheral inflammation [54, 56] and metabolic functions, including lipid metabolism [57], and such effects differ among patients with varying responses to these medications [45, 58]. Consequently, differences in blood markers have been observed between responders and non-responders [59].

In this study, we used five well-established behavioral paradigms to systematically assess three dimensions of EF in healthy participants and patients. Baseline assessments of these EF dimensions were used to identify core subtypes in schizophrenia from the main sample ($N = 160$) using two fuzzy clustering methods. These subtypes were validated for their replicability and classifiability in an independent "recurring local validation" sample ($N = 114$) recruited at a different time (Fig. 1). Afterward, the longitudinal stability of the subtypes and their relationship with remission status were evaluated in a re-assessed subset ($N = 86$) of the original main sample. A linear mixed model was employed to analyze remission outcomes in relation to changes in EF, symptoms, and peripheral blood assays (including immune and metabolic markers) from baseline to follow-up. Finally, baseline-determined subtypes were used to predict remission outcomes, indicating potential clinical applications.

Methods

Sample information

We recruited 195 patients with schizophrenia aged 35.35 ± 9.35 years (81 females), and 169 age- and sex-matched healthy individuals (HC group) who did not have a history of mental illness or brain injury. Among this patient cohort, 160 participants who completed EF assessments on five task paradigms and peripheral blood assays were included for present analyses (Table 1; details in Additional file 1: Supplementary Methods). This observational, case–control study was reported in accordance with the STROBE guidelines (see Additional file 2). The diagnosis of schizophrenia was established by two resident psychiatrists according to the International Classification of Diseases criteria (Tenth Revision) and confirmed using the Structured Clinical Interview for *Diagnostic and Statistical Manual of Mental Disorders* IV axis I disorders. Patients were recruited from the inpatient unit. Our cohort was not in a severe state at baseline; patients were clinically stable, having traversed the acute phase. Based on criteria by *Leucht et al.* (2005) [60] comparing the PANSS with clinical global impressions (CGI), approximately 80% of our patients were categorized as mildly-to-moderately ill (Additional file 1: Fig. S1), with no patients (0%) falling within the severely ill range (PANSS ≥ 120). Patients were monitored under naturalistic 'treatment-as-usual' conditions throughout the follow-up period to maximize ecological validity. While adjunctive therapies such as transcranial magnetic stimulation were excluded, attending psychiatrists retained full autonomy to make evidence-based adjustments to antipsychotic regimens. Most patients were on atypical antipsychotics (96.9%) with a few receiving typical (3.1%) in the main sample (Additional file 1: Table S2) [61]. Specific drug types were not restricted. Patients with a high olanzapine-equivalent dosage of > 35 mg/day, an evident extrapyramidal symptom that would in practice influence the conduct of behavioral tasks (Additional file 1: Fig. S3), and concurrent medication including those (e.g., benzodiazepines, antihistamines, anticholinergics, or sedatives) that may affect cognitive functions [62] were excluded

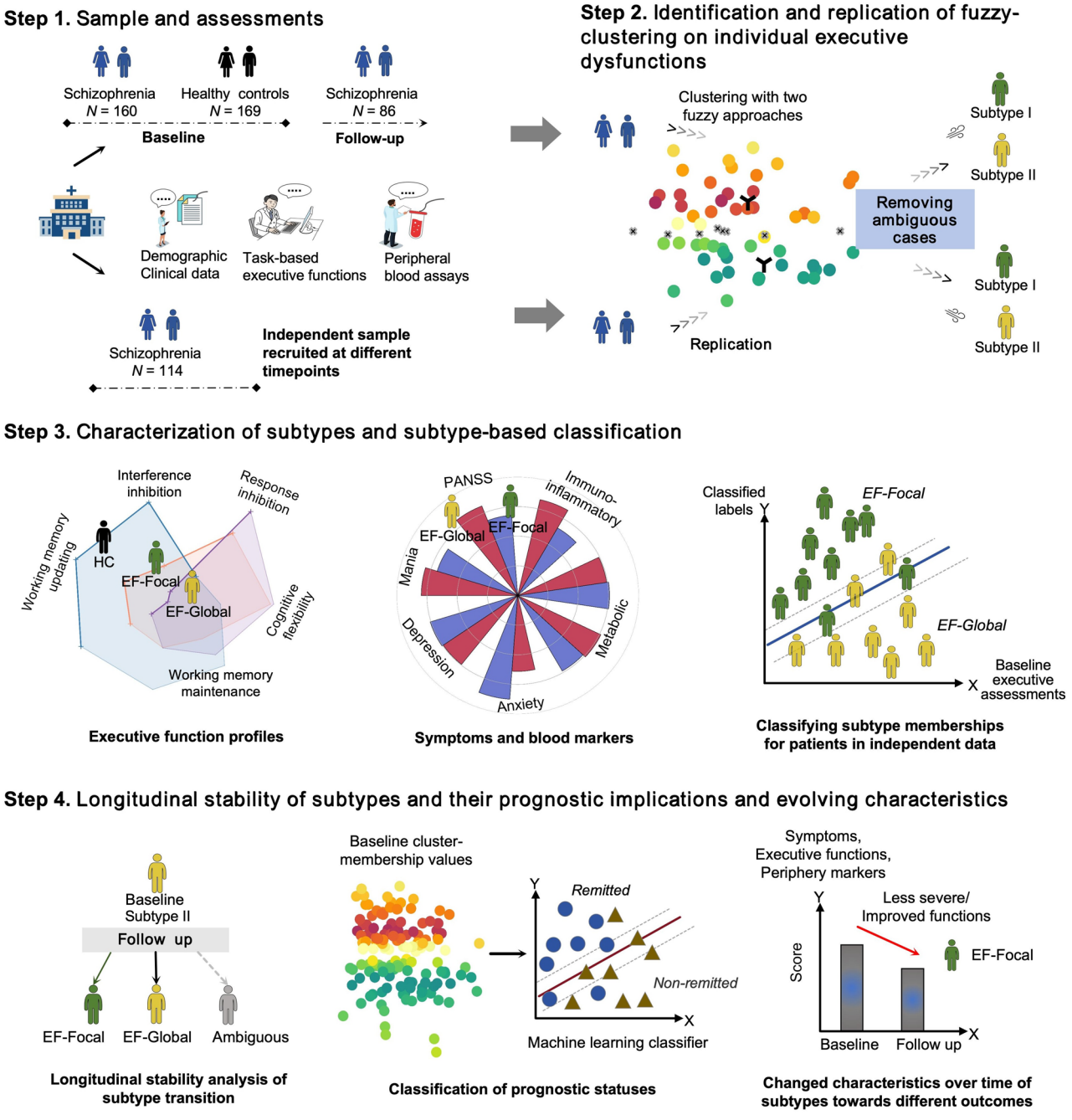


Fig. 1 Study overview

(cf. Additional file 1: Table S3). A few patients were on antipsychotic polypharmacy; there are multiple antipsychotic drugs involved. The doses of these antipsychotics were converted into olanzapine equivalent dosage as in our previous works [63, 64] and others [65, 66]. Apart from the dose of Perospirone which was converted following the Japanese dosing guidelines [67] as this drug is widely used in Asian populations, current drug dosages of other antipsychotics were olanzapine-equivalent transformed using equivalency ratios defined in *Gardner et al.* (2010) [68]; these ratios were derived based on consensus estimates of clinically equivalent doses from international survey of experienced and expert clinicians and researchers (Additional file 1: Table S4) [65, 66]. To test for the replicability and generalizability of identified EF subtypes, a local validation sample of schizophrenia with identical inclusion and exclusion criteria was collected independently 1 year after main sample collection ($N = 114$). Participants in the validation sample did not overlap with those in the main sample and completed the

Table 1 Demographic, clinical, and EF characteristics of participants

(M ± SD) / *n* (%)	Baseline sample		Independent sample		
	Schizophrenia (*N* = 160)	Healthy controls (*N* = 169)	Schizophrenia (*N* = 114)	Statistic value	p_{FDR}
Demographic variables					
Age	35.21 ± 9.53	37.69 ± 13.71	36.46 ± 9.65	−1.9133[a] 0.890[b] −1.057[c]	0.227[a] 0.746[b] 0.583[c]
Sex					
Male	97 (60.62%)	88 (52.07%)	75 (65.79%)	2.108[a] 4.699[b] 0.555[c]	0.234[a] 0.048[b] 0.730[c]
Female	63 (39.38%)	81 (47.93%)	39 (34.21%)		
Ethnicity, Han	142 (88.75%)	152 (89.94%)	93 (81.58%)	0.029[a] 3.407[b] 2.248[c]	0.987[a] 0.087[b] 0.682[c]
Education, years	10.88 ± 4.51	10.90 ± 3.94	10.71 ± 4.27	−0.032[a] 0.370[b] 0.319[c]	0.974[a] 0.746[b] 0.750[c]
BMI	23.63 ± 3.78	23.95 ± 4.32	24.58 ± 4.30	−0.713[a] −1.215[b] −1.900[c]	0.635[a] 0.746[b] 0.235[c]
Residence, urban	92 (57.50%)	98 (57.99%)	62 (54.39%)	< 0.001[a] 0.228[b] 0.151[c]	1.000[a] 0.633[b] 0.797[c]
SES	23.16 ± 7.61	23.88 ± 5.80	23.61 ± 7.73	−0.961[a] 0.325[b] −0.470[c]	0.635[a] 0.746[b] 0.750[c]
Employed, yes	40 (25.00%)	111 (65.68%)	38 (33.33%)	53.148[a] 27.290[b] 1.879[c]	< 0.001[a] < 0.001[b] 0.682[c]
Only child, yes	53 (33.12%)	24 (14.20%)	32 (28.07%)	15.380[a] 7.399[b] 0.576[c]	< 0.001[a] 0.017[b] 0.730[c]
Marital status					
Unmarried	96 (60.00%)	54 (31.95%)	65 (57.02%)	56.243[a] 37.524[b] 0.758[c]	< 0.001[a] < 0.001[b] 0.860[c]
Married	38 (23.75%)	108 (63.91%)	32 (28.07%)		
Divorced	25 (15.62%)	7 (4.14%)	16 (14.04%)		
Widowed	1 (0.62%)	0 (0.00%)	1 (0.88%)		
Smoking history					
Never	93 (58.13%)	122 (72.19%)	64 (56.14%)	7.183[a] 8.659[b] 1.206[c]	0.055[a] 0.026[b] 0.730[c]
1–3 years	16 (10.00%)	11 (6.51%)	8 (7.02%)		
> 3 years	51 (31.87%)	36 (21.30%)	42 (36.84%)		
Alcohol consumption history					
Never	107 (66.88%)	99 (58.58%)	70 (61.40%)	3.089[a] 4.646[b] 2.409[c]	0.285[a] 0.112[b] 0.730[c]
Occasionally	46 (28.75%)	64 (37.87%)	34 (29.82%)		
Regularly	7 (4.38%)	6 (3.55%)	10 (8.77%)		
Executive functions					
Interference inhibition (*Stroop task*)					
Reaction times in incongruent stimuli	757.24 ± 152.37	663.61 ± 149.02	736.82 ± 158.59	5.63[a] −3.90[b] 1.07[c]	< 0.001[a] < 0.001[b] 0.401[c]
Stroop interference effect	−20.33 ± 74.22	−7.04 ± 44.23	−26.41 ± 90.34	−1.96[a] 2.12[b] 0.59[c]	0.051[a] 0.035[b] 0.648[c]
Response inhibition (*Go/No – Go* task)					
Accuracy in No – Go trials	0.84 ± 0.12	0.88 ± 0.11	0.82 ± 0.13	−2.90[a] 4.08[b] 1.44[c]	0.005[a] < 0.001[b] 0.265[c]
Working memory updating (*Running memory* task)					

Table 1 (continued)

(M ± SD) / *n* (%)	Baseline sample		Independent sample		
	Schizophrenia (*N* = 160)	Healthy controls (*N* = 169)	Schizophrenia (*N* = 114)	Statistic value	p_{FDR}
Accuracy in 1,750 ms	0.58 ± 0.26	0.74 ± 0.21	0.52 ± 0.32	−6.01[a]	< 0.001[a]
				6.47[b]	< 0.001[b]
				1.76[c]	0.192[c]
Accuracy in 750 ms	0.50 ± 0.29	0.66 ± 0.22	0.44 ± 0.32	−5.66[a]	< 0.001[a]
				6.60[b]	< 0.001[b]
				1.74[c]	0.192[c]
Working memory maintenance (*Digit span backward* task)					
Span in digit span backward task	5.08 ± 1.52	6.06 ± 2.11	4.64 ± 1.58	−4.83[a]	< 0.001[a]
				6.45[b]	< 0.001[b]
				2.31[c]	0.152[c]
Cognitive flexibility (*Number switching* task)					
Switching cost	305.19 ± 295.26	198.95 ± 238.37	314.20 ± 303.06	3.58[a]	0.001[a]
				−3.41[b]	0.001[b]
				−0.25[c]	0.807[c]
Electronic medication records					
First episode, yes	6 (3.75%)	N/A	10 (8.77%)	2.21[c]	0.150[c]
Family history of psychiatric disorders, yes	28 (17.50%)	N/A	12 (10.53%)	2.07[c]	0.150[c]
Frequency of episodes	6.18 ± 4.30	N/A	5.00 ± 3.34	2.55[c]	0.045[c]
Age at onset	27.25 ± 8.42	N/A	25.94 ± 8.51	1.26[c]	0.209[c]
Duration of disorder	9.78 ± 7.02	N/A	11.40 ± 7.85	−1.76[c]	0.160[c]
Olanzapine-equivalent dose (mg/day)	14.79 ± 6.48	N/A	13.68 ± 5.66	1.51[c]	0.177[c]
PANSS Scale					
PANSS Positive subscale	21.66 ± 4.53	N/A	20.58 ± 8.06	1.30[c]	0.268[c]
PANSS Negative subscale	21.19 ± 6.83	N/A	23.19 ± 9.50	−1.92[c]	0.156[c]
PANSS General Psychopathology subscale	39.73 ± 7.09	N/A	37.89 ± 13.23	1.35[c]	0.267[c]
PANSS Total score	82.58 ± 14.46	N/A	81.66 ± 27.78	0.32[c]	0.799[c]

Note. Superscript letters denote between-group comparisons—[a] Main-sample schizophrenia vs. Main-sample healthy control; [b] Main-sample healthy control vs. Independent-sample schizophrenia; [c] Main-sample schizophrenia vs. Independent-sample schizophrenia. ***Abbreviations***. N/A, not applicable; PANSS, Positive and Negative Syndrome Scale; SES, socioeconomic status; BMI, body mass index

same assessments. This study was approved by the ethics committees of the Third People's Hospital of Lanzhou City (approval no. TPHL202303) and the Northwest Normal University (approval no. NWNU202404), Lanzhou, China. Written informed consent from all participants was obtained.

In the main sample, 86 patients were followed after a 4–6-week interval. Such a follow-up period has been commonly implemented in literature [44, 69], which should well capture early treatment response (e.g., PANSS reduction and changes in inflammatory markers) [70, 71] while balancing the drop rates. Among these patients, 7 (8.1%) switched antipsychotic medications, while 22 (25.6%) underwent dose titrations with a mean olanzapine-equivalent dosage increase of 21.4% (Additional file 1: Table S17). There is an active, multicomponent therapeutic environment in the inpatient setting. It encompasses structured daily routines, daily physical exercise, group psychoeducation (1–2 sessions/week), occupational activities (2–3 sessions/week), and wellness activities. These ward-based psychosocial interventions work synergistically with pharmacological treatments to drive continuous improvement following initial stabilization [72, 73]. Previous research has suggested that antipsychotic response trajectories often extend well beyond initial stabilization [74, 75]. Remission versus non-remission prognostic status was determined using the Remission in Schizophrenia Working Group (RSWG) [76], i.e., using scores of ≤ 3 points on PANSS items (P1, P2, P3, N1, N4, N6, G5, G9). Fifty-eight participants at follow-up were identified as being in remission; the other 28 participants were not in remission. Considering a lack of consensus, we used an alternative definition of remission defined by reductions in the total PANSS score from baseline to follow-up, as follows: (1) a 25% reduction (68 remitted vs. 18 non-remitted); (2) a 35% reduction (48 remitted vs. 38 non-remitted); and (3) a 50% reduction (16 remitted vs. 70 non-remitted). While the RSWG criteria focused on specific PANSS items, percentage reductions in total PANSS score provided an evaluation on all items. This systematic assessment was to ensure that our findings were not driven by idiosyncrasies in how we defined treatment response and helps establish the robustness of our subtyping in indicating remission status [77, 78]. The use of different criteria to define remission can naturally result in a few

inconsistent assignments of patients to the remission or non-remission groups (cf. Additional file 1: Fig. S4A), but allows us to better appreciate to which degree the percentage reduction cutoff would best suit our patient samples as they go along with the RSWG definitions as well as the related treatment responses. Symptom severity in each patient with schizophrenia was evaluated using the PANSS [79]. Scores for four symptom dimensions (i.e., positive, negative, affective, and cognitive factors) were derived for each patient using the Dimensions and Clustering Tool for Schizophrenia Symptomatology (DCTS; http://webtools.inm7.de/sczDCTS/) [63, 64, 80–82]. Specifically, an individual's score for each dimension was determined by calculating the dot product of the coefficients loaded on that dimension and the 30 PANSS item scores of this individual. Nevertheless, each dimension is driven (mainly) by a few items: positive (P1 "delusions," P3 "hallucinations," and G9 "unusual thought content"), negative (N1 "blunted affect," N2 "emotional withdrawal," and N4 "apathetic social withdrawal"), affective (G2 "anxiety," G6 "depression," and G4 "tension"), cognitive (P2 "conceptual disorganization," N5 "difficulty in abstract thinking," G12 "lack of judgment and insight," and G11 "poor attention"); these are with relatively high coefficients loaded on a particular dimension (a column in Additional file 1: Fig. S2 (each value represents a factor) within the a priori 4-factor model). Higher scores denote more severe symptoms within each dimension. Depressive, anxiety, and manic symptoms were further assessed using the Hamilton Depression Rating Scale (HAMD), the Hamilton Anxiety Rating Scale (HAMA), and the Bech–Rafaelsen Mania Scale (BRMS) [83–85].

EF assessments

Seven metrics from five behavioral tasks were employed to assess three EF dimensions [86], as previously proposed: cognitive flexibility, inhibition, and working memory (Additional file 1: Table S5). The inhibition dimension included response inhibition and interference inhibition, while the working memory dimension consisted of maintenance and updating [87]. Interference inhibition was measured using the Stroop Color–Word Task (*Reaction Time for Incongruent Stimuli and Stroop Interference Effect*) [88], response inhibition was measured using the Go/No-Go Task (*Accuracy in No-Go Trials*) [89, 90], cognitive flexibility was measured using the Number-switching Task (*Switching Cost*) [91], working memory updating was measured using the Running Memory Task (*Accuracy at 1750 and 750 ms, respectively*), and working memory maintenance was measured using the Digit Span Backward Task [92, 93] (see Additional file 1: Table S5 and details in Supplementary Methods) [94–96].

Peripheral immune and metabolic marker assays

Eight immunoinflammatory indices were assessed. Our selection of peripheral blood markers was grounded in the "immune hypothesis" of schizophrenia [97–99]. There is ample evidence that innate immune activation is closely tied to the disorder's pathophysiology [100–102]. We also assessed seven metabolic indices, as they carry significant implications for treatment response in schizophrenia [103, 104]. Elevated levels of triglycerides, total cholesterol, low-density lipoprotein cholesterol, lactate dehydrogenase, creatine kinase, and the ratio of aspartate aminotransferase to alanine aminotransferase have been associated with poorer metabolic function, while higher high-density lipoprotein cholesterol (HDL-C) levels are associated with more favorable metabolic health [105]. All biomarkers were measured using standardized clinical laboratory methods (detailed in Additional File 1: Supplementary Methods). Blood samples were collected from the patients with schizophrenia as part of their routine clinical assessment, but not from healthy controls.

Identification and replication of subtypes using baseline EF assessments

EF scores in the patient group were normalized by the mean and standard deviation values of those assessed in the healthy group and adjusted for age, sex, and education level, using the regression weights evaluated within the healthy population. Then, residuals of EF assessments were used for clustering patients into cognitive subtypes. We applied a fuzzy c-means (FCM) clustering technique, which provided cluster membership degrees for all patients (detailed in Additional File 1: Supplementary Methods) [106]. The optimal cluster number was determined based on the fuzzy silhouette index, the Xie and Beni index, and Kwon's index (Additional file 1: Table S6) [107, 108]. As the clustering quality for each number of clusters would be affected by the fuzzifier m (i.e., the exponent for the fuzzy partition matrix [U]; $1 < m < \infty$) parameter, which controls the amount of fuzzy overlap between clusters, we tuned this parameter to determine the optimal value of m from 1.5 to 2.5 with a step length of 0.25 and that of cluster number from 2 to 7 [106, 109], based on their achievable clustering quality according to the three evaluation indices. Variability was assessed by the change of values of the three evaluation indices with m values for a particular cluster number; a smaller change refers to better stability, irrespective of the degree of overlaps between clusters. Stability was tested by bootstrap resampling (detailed in Additional File 1: Supplementary Methods) [110, 111].

Given the heterogeneous nature of schizophrenia and the observation of multiple patients with ambiguous memberships, a cutoff over the membership likelihoods was adopted to remove cluster-ambiguous patients. For

this, additional evidence from Gaussian mixture modeling (GMM) was considered (detailed in Additional File 1: Supplementary Methods). Specifically, patients were clustered again using GMM, and the optimal cluster number was determined by the Bayesian information criterion. After assigning patients to the clusters, we took the intersection of the c-means and GMM results. Since patients within the intersection set should have a higher membership degree than those outside, we followed the heuristic idea in our previous work [63] to define a cutoff that effectively discriminates "inside" and "outside" patients while considering the sample size retained for subsequent analyses. Specifically, the c-means membership values of 'inside' patients belonging to each cluster were compared with the remaining individuals; a cutoff value thus would facilitate the identification of cluster cores as the "core subtypes" of schizophrenia.

The same clustering approach (i.e., both c-means and GMM) was repeated on the independent "recurring local validation" sample containing 114 patients with schizophrenia.

Characterization of the yielded subtypes

To characterize the "core" EF subtypes following an exclusion of ambiguous cases, one-way analysis of covariance was employed to investigate differences in EF assessments, symptoms, and peripheral blood assays while controlling the effects of sociodemographic variables, including age, sex, and years of education (Fig. 1). Comparisons regarding EF involved both the subtypes and the HC group, while symptoms and blood assays were available for patients. Resulting p values were corrected for multiple comparisons using the false discovery rate (FDR) method [112].

Subtype-based classification

Furthermore, we tested for whether patients can be accurately classified per subtype using EF assessments (Fig. 1). Essentially, we trained classifiers (random forest) with nested cross-validation (repeated 100 times) on the baseline EF data to discriminate the cluster label of each patient. The optimal model derived from the cross-validation runs [113–118], without any further tuning, was applied on the EF data assessed in the independent validation sample once; the predicted labels were then compared with those obtained when re-clustering the validation sample. Performance of the classifier was assessed using area under the curve (AUC), balanced accuracy (BAC), sensitivity, and specificity. In the classifier setting, Subtype II was defined as the positive category, while Subtype I was defined as the negative category. Therefore, "sensitivity" denotes the model's capacity for detecting true positives (Subtype II cases), while "specificity" reflects the ability to accurately identify Subtype I cases. We moreover trained the classifier by additionally including the healthy subject group [119] within the baseline sample (i.e., multi-class classification) and tested it on an independent sample of patients. This approach aimed to determine which subtype was more likely to be misclassified as healthy and which was more distinct from the healthy population in terms of EF assessments.

Longitudinal stability of EF subtypes

The EF characteristics of the follow-up sample ($N = 86$, with a follow-up period of 4–6 weeks) were firstly mapped onto the cluster centers from the baseline sample (Fig. 1). Specifically, EF re-assessments at follow-up were z-score–standardized by those assessed in healthy subjects at baseline, and the covariates of age, sex and educational level were adjusted using the regression weights estimated on the baseline EF data. Based on the ensuing residuals, the squared Euclidean distance of each patient to the cluster center derived from baseline EF assessments was calculated. Based on the distance metric, membership values were derived for each participant. The fuzziness coefficient was maintained at the same level as in the baseline ($m = 2.5$). The ensuing membership values were used to assign each participant to a specific cluster, if this cluster, compared to others, had a higher membership value tied to this participant. Patients with a value below the cutoff of 0.6 were filtered out, resulting in two core subtypes in the follow-up data. Subtype labels were compared between baseline and follow-up for the same patients to assess their longitudinal stability. Patients assigned to the same clusters in both initial and follow-up stages were regarded as longitudinally stable.

The relationship of subtypes with remission outcomes

We used binary logistic regression analysis to investigate whether the two EF subtypes differed in remission outcomes (Fig. 1). In a logistic regression model, remission status (remission vs. non-remission) served as the dependent variable, with EF subtype acting as the predictor of interest. Effects of sociodemographic variables (age, sex, and educational level) and medication (olanzapine-equivalent dosage) were adjusted in the logistic model [120]. Remission outcomes of patients were assessed according to the RSWG criteria [76]. Additional analyses were performed using alternative definitions of a remission based on a reduction in total PANSS score.

Remission statuses classification based on baseline subtype statuses of patients

Baseline c-means membership values were used as input features to train a machine learning classifier (see Additional File 1: Supplementary Methods) aimed at determining patients' remission status (remission versus

non-remission, based on the RSWG criteria) [77, 114, 121–125]. This approach allowed us to evaluate whether the model tended to classify patients with high Subtype I or Subtype II membership value as remission or non-remission. The trained classifier was moreover applied to the patients who were lost to follow-up, which helped test if the discriminated remission conditions were in a ratio similar to that observed in reassessed patients.

Change in characteristics over time associated with each subtype toward different prognostic statuses

We moreover explored whether patients belonging to different subtypes presented varied changes in EFs, symptoms, and blood immune and metabolic indices from baseline to follow-up. This analysis was conducted using multiple linear mixed-effects models [126], incorporating subtype and time point (baseline versus follow-up) as fixed effects. The dependent variables included (1) scores in four dimensions of psychopathology, (2) EF assessments, and (3) peripheral blood immune and metabolic markers. Sociodemographic variables of age, sex, and years of education were included as covariates, and their effects were adjusted in the model. An interaction effect between time and subtype suggests that the relationship between the dependent variable and one subtype differs from that with the other over time. The resulting statistics were adjusted for multiple comparisons using the false discovery rate method. In addition, when a significant interaction effect occurred regarding peripheral blood markers, we assessed whether their changes would mediate an association between EF subtypes identified at baseline and the symptom-load and EF performance when reassessed through ordinary least squares path analysis of regression [127]. A percentile bootstrapping approach was employed to derive the CI of the indirect effect of possible mediators on subtype of EF/symptom associations. This included repeatedly resampling the original data for 10,000 times and performing the same regression analysis on the resampled data; if the 95% CI of the generated statistics did not cross zero, it indicated an indirect effect for a tested mediator.

Complementary analyses

Several analyses were supplemented: (1) While the use of fuzzy clustering approach better aligns with the heterogeneous psychopathology in schizophrenia, we applied *k*-means clustering, a hard-clustering method, to derive binary subgroups, for a more direct comparison. (2) While we adopted a membership value of 0.6 to filter out patients with an ambiguous assignment in our main analyses, we repeated the study with different cutoffs (0.55, 0.65, 0.70) to determine core patient subtypes and compared their EF, clinical symptoms, inflammatory levels, and treatment responses. (3) Given the significant differences in baseline overall symptom severity between the two EF-defined subtypes, we carried out additional analyses to determine whether the differing prognostic outcomes were simply a reflection of baseline symptom severity levels of individual patients. To investigate this, we analyzed the correlation between baseline total PANSS scores and EF assessments. Next, we adjusted for baseline total PANSS scores in our logistic regression analysis to determine if the remission outcomes for each subtype stayed consistent when overall symptom severity wasn't adjusted for, as in our original analysis. We also assessed diagnostic and prognostic classification based on subtypes derived from EF residuals, additionally adjusting the total PANSS score. (4) To further evaluate whether antipsychotic usages would have impacted our EF-based subtyping, we conducted two control analyses: (1) regressing out the effect of antipsychotic medication (olanzapine-equivalent dosage) on baseline EF assessments using general linear models, and then repeating the clustering procedure; (2) excluding patients treated with typical antipsychotic drugs and those who presented moderate extrapyramidal symptoms (a total of 8 patients). 5) The current naturalistic observational study incorporated variance in test-retest intervals (median: 35 days), and we additionally assessed its impact on EF assessments, subtyping, as well as the correlation and classification of remission status. The temporal robustness of our machine learning framework was also evaluated on the training set comprising patients with assessments within 35 days ($n = 44$) and an independent testing set comprising those assessed after 35 days ($n = 42$).

Results

Two replicable core EF subtypes were identified

FCM clustering performed on the adjusted and normalized EF assessments revealed an optimal two-cluster solution (Fig. 2A and B and Additional file 1: Fig. S5). That is, no matter which value of the fuzzifier coefficient m was selected, the fuzzy silhouette index reached its highest value while both of the Xie-Beni and Kwon's V indices achieved their lowest values at the cluster number of 2. For clustering with two partitions, the best cluster quality was revealed when m was set to 2.5 (Additional file 1: Fig. S5). FCM results were replicated by using a GMM, which demonstrated likewise an optimal two-cluster solution in the same data (Fig. 2C). The c-means membership values of patients within the intersection set for each cluster were significantly greater than those outside the cluster and the intersection set defined by c-means and GMM ($p < 0.01$, Wilcoxon rank-sum test) (Fig. 2D and E and Additional file 1: Table S8). In this context, a heuristic cutoff at 0.6 was chosen, which effectively discriminated patients inside the intersection set for each cluster from

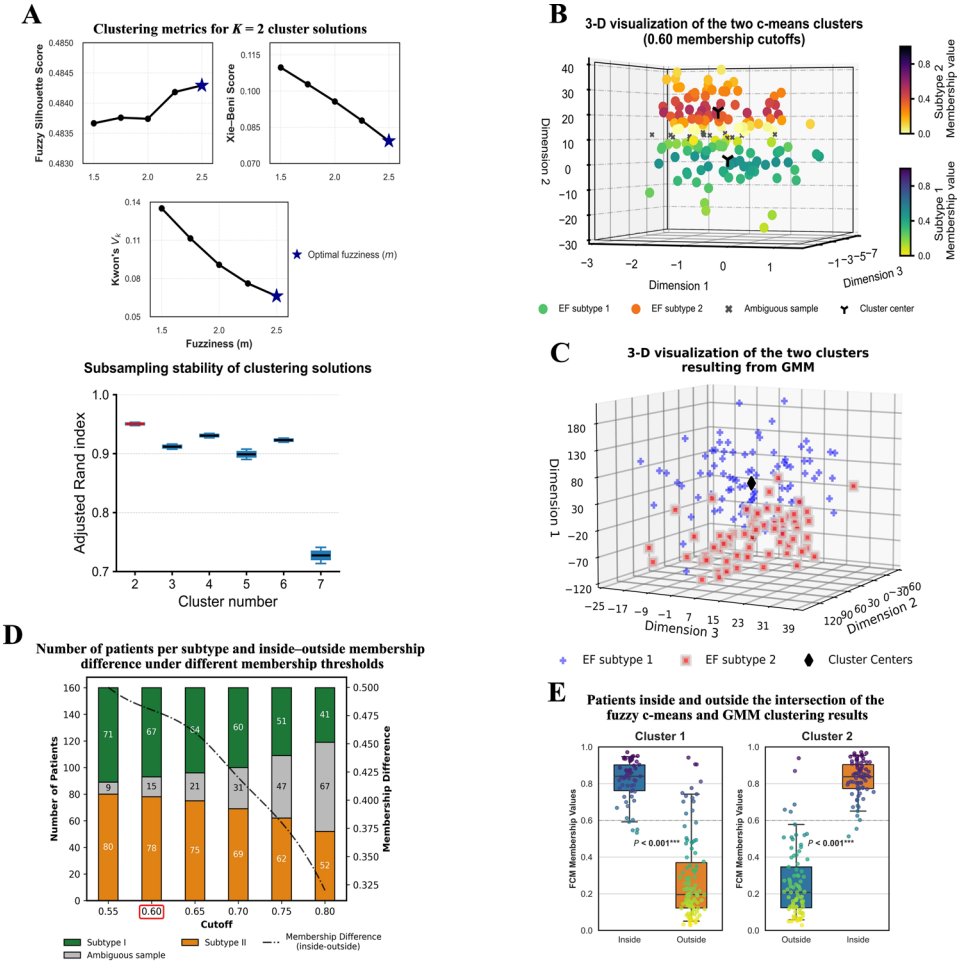


Fig. 2 FCM clustering on baseline EF data. **(A)** FCM clustering results for EF features in the baseline sample. This included clustering quality indices and subsampling (85%) stability assessment repeated for 1,000 times, with higher adjusted Rand index values indicating greater stability. **(B)** The sample distribution results of schizophrenia patients ($N=160$) in the FCM clustering model ($k=2$, $m=2.5$). Clustering was based on seven EF indicators. To facilitate visualization, principal component analysis was used to reduce the dimensionality of these indicators, and we took the loading values on the first three components following the principal component analysis. The FCM membership values of each sample are color-mapped in the scatterplot. The black "x" markers between the two subtypes represent ambiguous samples ($N=15$). **(C)** The optimal clustering result of the GMM for EF input values. The blue "+" in the plot represents the sample distribution of EF Subtype I, while the red "+" represents the sample distribution of EF Subtype II. **(D)** The number of patients in Subtype I, Subtype II, and ambiguous class under different membership thresholds. The curve represents the membership difference values (inside − outside). **(E)** The distribution of inside and outside sample FCM membership values for clusters 1 and 2. The inside degree is the c-means membership values of patients assigned to the same cluster by GMM and FCM, while other patients are outside. *$p<0.05$, **$p<0.01$, ***$p<0.001$

those outside to choose the cluster cores. As a result, two "core" subtypes for schizophrenia of 78 and 67 patients, respectively, were defined after filtering out 15 ambiguous cases (7 and 8 in each original cluster) (Fig. 2D and Additional file 1: Table S8). This dichotomous differentiation was replicated in the independent sample, which likewise supported a cutoff value of 0.6 that effectively discriminated patients within the c-means GMM intersection set for each cluster and those outside (Fig. 3A and B). Using 0.6 as the cutoff value, the proportion of ambiguous cases approached closest between the results derived from the baseline (9%) and the independent local validation (6%) samples, while ambiguous cases grew prominently, especially in the validation sample, when more conservative cutoffs were applied (25% using 0.65, 43% using 0.7) (Fig. 3C). Of note, among the 74 patients lost to follow-up, a similar distribution of cluster memberships was observed ($p>0.35$) (Additional file 1: Fig. S6A). The ratio of patients in each subtype (48.6% EF-Focal, 41.9% EF-Global) and that in the ambiguous class (9.4%) did not differ significantly from those of patients who were reassessed (36% EF-Focal, 54.6% EF-Global; $\chi^2=2.84$, $p=0.242$). This indicates that our reassessed patients effectively represented the EF heterogeneity of those lost to follow-up.

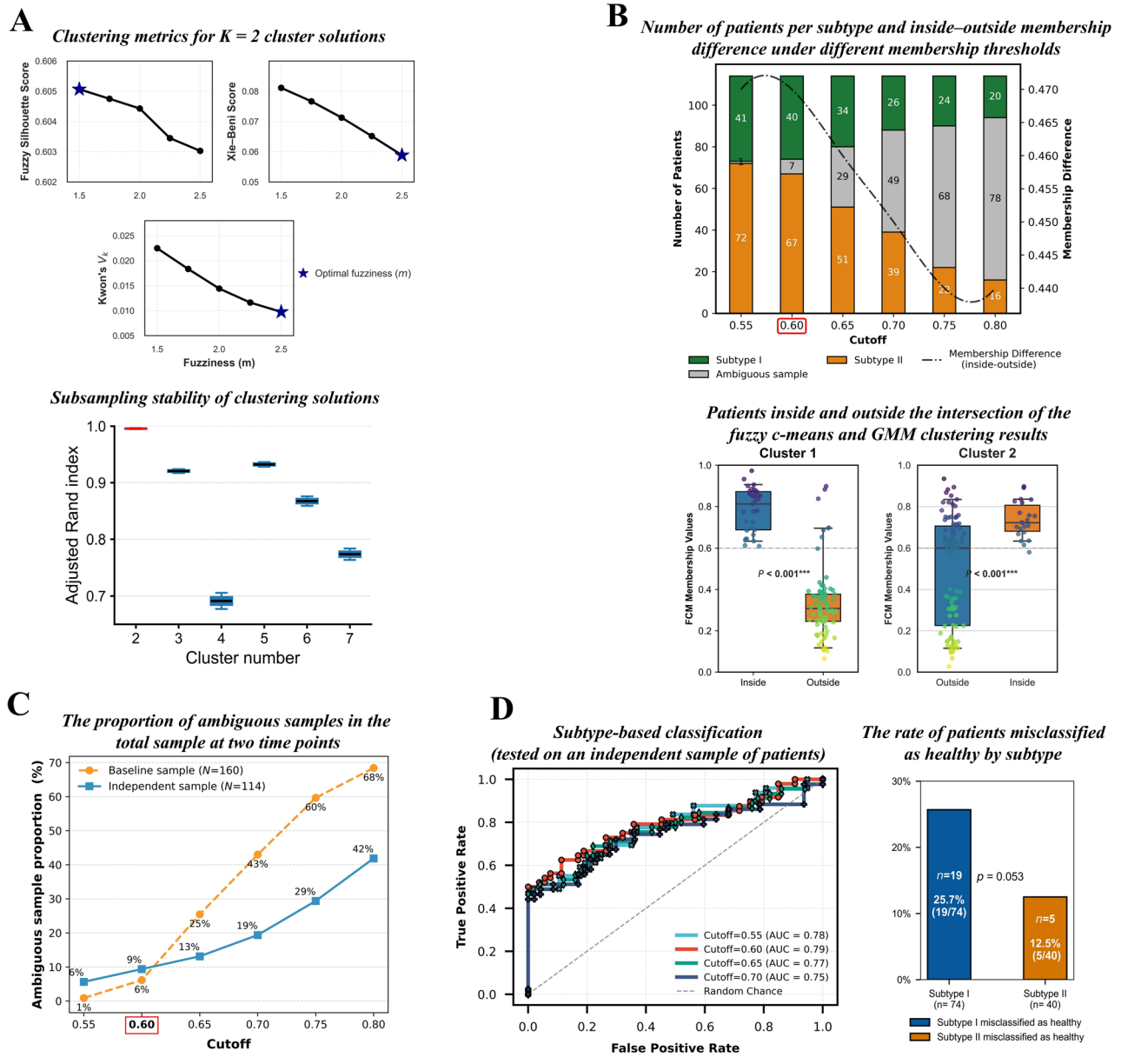


Fig. 3 Recurring local validation. **(A)** shows the results of re-clustering EF features in the validation sample ($N = 114$) using FCM, with the clustering procedure consistent with the baseline sample. This includes clustering quality indices and subsampling (85%) stability assessment repeated for 1,000 times, with higher adjusted Rand index values indicating higher stability. **(B)** The upper plot displays the quantity distribution of "inside" and "outside" samples under different membership thresholds, with the curve representing membership difference values (inside – outside). The lower plot shows the distribution of inside and outside sample FCM membership values for clusters 1 and 2. The inside degree is the c-means membership values of patients assigned to the same cluster by GMM and FCM, while other patients are outside. **(C)** shows the change in the proportion of ambiguous samples in the total sample as the membership cutoff varies at two time points (baseline vs. validation sample recruited at a separate time). **(D)** Classifiers were trained based on baseline EF assessments and used for determining the subtype labels for patients in the validation sample. The true subtype labels of the validation sample were obtained by independently re-clustering using the same parameter procedure as the main sample. Classifiers were also trained on the pooled categories (Subtype I, Subtype II, HC). In the validation sample, patients in Subtype I were more frequently misclassified as healthy than those in Subtype II. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$

Subtypes were characterized by varying profiles in EF, symptomatology, and immunometabolics

EF

The comparison of EF assessments between each of the subtypes and the healthy control group demonstrated that the patients in Subtype I presented relatively preserved EF with deficits restricting to only two variables (EF-Focal) while those in Subtype II showed broad EF impairments (EF-Global). Specifically, except for one variable (Stroop interference effect), which assessed the

interference inhibition dimension ($p_{FDR=}$0.86, $d = 0.064$), the cognitive flexibility (switching cost, $p_{FDR=}$0.021, $d = -0.396$), response inhibition (accuracy in *No-Go trials*, p_{FDR}=0.049, $d = 0.358$), interference inhibition (variable reaction times [RTs] for incongruent stimuli, $p_{FDR} < 0.001$, $d = -0.569$), working memory updating (*variable accuracy in 1750 ms*: $p_{FDR=}$0.004, $d = 0.481$; *variable accuracy in 750 ms*: $p_{FDR} < 0.001$, $d = 0.556$), and maintenance ($p_{FDR=}$0.024, $d = 0.445$) were significantly different between EF-Global and healthy subjects (Fig. 4A and Additional file 1: Table S9). Compared to healthy subjects, EF-Focal patients showed greater restrictions in certain EF deficits, i.e., only response inhibition ($p_{FDR=}$0.049, $d = -0.354$) and cognitive flexibility ($p_{FDR=}$0.021, $d = -0.415$). Meanwhile, EF-Focal and EF-Global differed significantly in working memory updating (*variable accuracy in 750 ms*: $p_{FDR=}$0.036, $d = 0.386$). Importantly, there are no significant differences in the rates of medication adjustments (no change, dose titration, or antipsychotic switching) between the two EF subtypes (χ^2 test, $p = 0.92$).

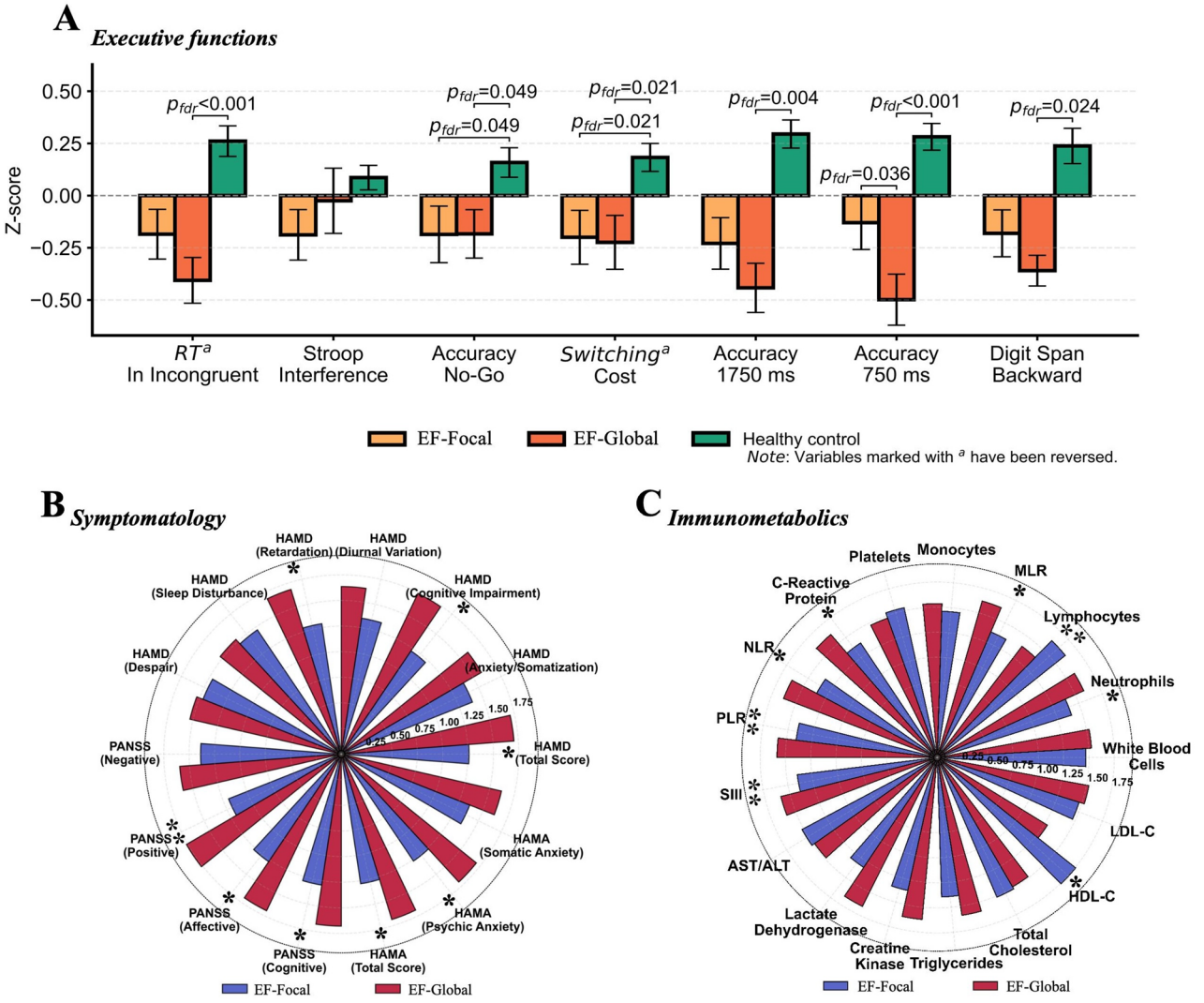


Fig. 4 Comparison of EF assessments, clinical symptoms, and peripheral blood markers between subtypes. One-way analysis of covariance was performed to test potential differences between the subtypes in **(A)** EF assessments, **(B)** clinical symptoms, and **(C)** peripheral blood markers (inflammation and metabolic), controlling for socio-demographic variables (age, sex, and years of education). All results were corrected for multiple comparisons using the false discovery rate method. *Abbreviation*: SCZ, schizophrenia; HAMD, Hamilton Depression Rating Scale; HAMA, Hamilton Anxiety Rating Scale; PANSS, Positive and Negative Syndrome Scale; WBC, white blood cell; NEUT, neutrophil; LYMPH, lymphocyte; MONO, monocyte; PLT, platelet; NLR, neutrophil-to-lymphocyte ratio; PLR, platelet-to-lymphocyte ratio; MLR, monocyte-to-lymphocyte ratio; SII, systemic immune-inflammation index; CRP, C-reactive protein; HDL-C, high-density lipoprotein cholesterol; LDH, lactate dehydrogenase; CK, creatine kinase; AST, aspartate aminotransferase; ALT, alanine aminotransferase. $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Clinical scales

EF-Global patients showed greater severity in their multidimensional symptom profiles, with significantly greater scores on three dimensions of PANSS assessments (positive symptoms: p_{FDR}=0.002, affective symptoms: $p_{FDR=}$0.020, and cognitive symptoms: $p_{FDR=}$0.026), total PANSS score (p_{FDR}=0.011), total HAMD score (p_{FDR}=0.024), HAMD cognitive impairment factor (p_{FDR}=0.011), and HAMD retardation factor (p_{FDR}=0.040) as well as the total HAMA score (p_{FDR}=0.020) and HAMA psychic anxiety factor (p_{FDR}=0.020), along with a non-significantly higher baseline OZP dosage ($p = 0.09$), compared to EF-Focal patients (Fig. 4B and Additional file 1: Table S10). Medication characteristics (type, dosage, concomitant use) did not differ significantly between EF subtypes (all $p_{FDR} > 0.155$; Additional file 1: Table S2), and none of the 28 tested associations between medication variables and EF performance reached significance (all $p_{FDR} > 0.521$; Additional file 1: Table S11).

Peripheral blood assessments

Regarding peripheral blood immune indices, patients in the EF-Global subtype exhibited a significantly higher neutrophil percentage ($p_{FDR=}$0.015), C-reactive protein levels ($p_{FDR=}$0.022), neutrophil-to-lymphocyte ratio (NLR) ($p_{FDR=}$0.017), platelet-to-lymphocyte ratio (PLR) ($p_{FDR=}$0.008), monocyte-to-lymphocyte ratio (MLR) (p_{FDR}=0.041), and the systemic immuno-inflammation index ($p_{FDR=}$0.008) compared to EF-Focal patients (Fig. 4C and Additional file 1: Table S10). In contrast, the lymphocyte percentage ($p_{FDR=}$0.009) was significantly lower in EF-Global patients. For peripheral metabolic indices, a significant between-subtype difference was only revealed in the HDL-C levels, which were lower in EF-Global patients ($p_{FDR=}$0.028). Although peripheral blood samples were not obtained from the healthy control group, there are reference ranges for the inflammatory and metabolic markers we assessed following laboratory hematological assays for the healthy populations in China (cf. Additional file 1 for details). Notably, relative to the upper bound of the reference range, patients in EF-Global presented a 42% elevation in CRP concentration on average, while the level of HDL-C reached closely to the lower bound of 1.04 mmol/L. In contrast, EF-Focal patients demonstrated generally better metabolic functions and lower levels of inflammation with the CRP level just around the reference upper bound of 5 mg/L.

Subtypes can be accurately classified, with more patients in the (milder) EF-Focal seeming healthy

Training a classifier within the baseline sample and applying it to the EF assessments of individuals in the validation sample yielded a high discriminative power of the two subtypes (AUC = 0.79, BAC = 0.72, sensitivity = 0.79, specificity = 0.64) (Fig. 3D, Additional file 1: Fig. S7 and Additional file 1: Table S12). Repeating the classification analyses on subtypes derived by alternative cutoff values in each patient's membership degree yielded inferior performance (Additional file 1: Table S12). Including HC when training the classifier misclassified more EF-Focal patients (19/74) in the validation sample as healthy than those in EF-Global (5/40) (Fig. 3D).

Subtypes were longitudinally stable

Among the 86 patients with repeated EF assessments, 78 patients were identified as exhibiting the core subtypes, while eight demonstrated an ambiguous cluster membership (highest membership degree < 0.6). The longitudinal stability analysis showed that more than 90% of the reassessed patients retained their subtype, with three patients each from EF-Focal and EF-Global shifted to the ambiguous cluster at follow-up (Fig. 5A). One ambiguous case at baseline was assigned to EF-Focal at follow-up, while three were assigned to EF-Global. Longitudinal stability decreased when more stringent cutoffs were used over the membership values, i.e., 80% for 0.65 and 74% for 0.7 (Additional file 1: Fig. S8).

Patients of different subtypes differed in remission outcomes that can be moreover predicted from baseline EF performance

The regression analysis, adjusting for baseline OZP dosage, revealed that EF subtype was a significant predictor of remission status when using RSWG criteria ($p = 0.013$) (Fig. 5C, Additional file 1: Fig. S9). Specifically, patients with EF-Global were significantly less likely to achieve remission compared to those with EF-Focal (odds ratio = 0.203, 95% confidence interval [CI] [0.058, 0.714]). The robustness of these results was confirmed through sensitivity analyses after excluding the 7 patients who underwent antipsychotic switching (odds ratio = 0.17, $p = 0.012$), using alternative remission criteria based on different total PANSS score reduction thresholds (35% [odds ratio = 0.302, $p = 0.030$] and 50% [odds ratio = 0.127, $p = 0.016$]) as well as just taking the patients whose remission or non-remission statuses were consistently defined by both RSWG criteria and a certain reduction ratio in total PANSS score (35%, as at this cutoff, percentage-reduction method achieved the highest degree of overlapped patients with the RSWG criteria; odds ratio = 0.202, $p = 0.020$) (Additional file 1: Fig. S4A) [78]. Taking only these consistently defined remission vs. non-remission patients when training our models on baseline FCM membership values to classify these two outcome types yielded a discriminative performance of AUC = 0.72 (Additional file 1: Fig. S4B–C).

The classifier achieved an AUC value of 0.78 in the follow-up subset sample, which indicated that the

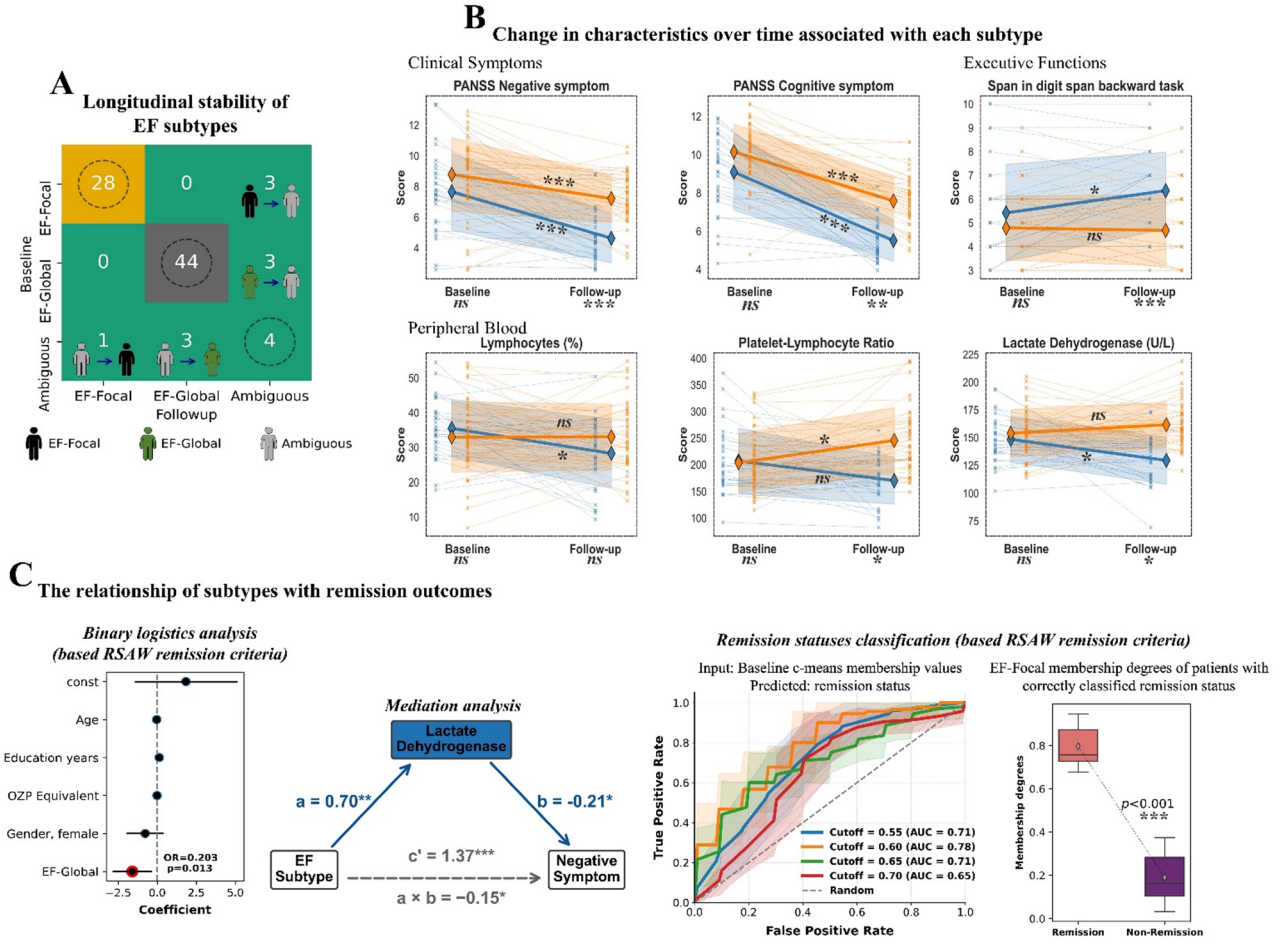


Fig. 5 Longitudinal stability and change of characteristics over time associated with each subtype toward different prognostic statuses. **(A)** shows the longitudinal stability results of EF subtypes under different FCM membership cutoff conditions. The EF subtype labels of the follow-up sample were calculated by mapping the EF features at the follow-up stage to the cluster centers of the optimal model from the baseline sample. The consistency between the subtype labels of the follow-up sample and the baseline sample was evaluated through consistency calculations. Black circles indicate stable individuals (i.e., subtype labels remained consistent between baseline and follow-up stages). **(B)** presents the analysis of changes in characteristics over time, showing the difference test results between the two EF subtypes in (1) clinical scales, (2) EFs, and (3) peripheral blood. Each line represents a patient. Blue line represents EF-Focal; orange line represents EF-Global. The significance markers on the x-axis indicate differences between baseline and follow-up groups (controlling for sociodemographic variables, including age, sex, and years of education). The significance markers on the lines indicate within-group differences between EF-Focal and EF-Global (controlling for sociodemographic variables). All analyses were adjusted for demographic variables (age, sex, and educational level). **C)** Binary logistic regression analysis examined whether one EF subtype was more likely to achieve remission after treatment. Significance was determined based on odds ratios. Mediation analysis was performed with path a representing the effect of EF subtypes on the change of lactate dehydrogenase over time. Path b tested the influence of this mediator on negative symptoms assessed at follow-up. c′ denotes the direct effect, while a × b represents the indirect (mediation) effect. Baseline FCM membership values were used to classify the patients' remission outcomes (remission vs. non-remission, according to RSWG criteria). The box plot on the right panel shows the FCM membership values for all correctly classified patients with respect to the EF-Focal subtype per their remission or non-remission condition. *ns*, no significant difference. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

model was more inclined to classify individuals with higher baseline EF-Focal membership as in remission and those with higher baseline EF-Global membership as in non-remission (Fig. 5C). The classification performance was comparable when using a PANSS 50% (AUC = 0.76) reduction criterion but was lower when PANSS 25% (AUC = 0.66) or 35% (AUC = 0.65) thresholds were applied (Fig. 5C and Additional file 1: Table S12). Among the cases accurately classified by the model, patients assigned to the remission group had significantly higher baseline EF-Focal membership values (mean: 0.80) than those classified as non-remission cases (mean: 0.19) ($p < 0.001$) (Fig. 5C). In addition, applying the classifier on the baseline memberships of unfollowed patients resulted in the proportion of remitted and non-remitted conditions being similar to the observed conditions in followed patients (χ^2=0.023, $p = 0.879$ for EF-Focal, χ^2=0.078, $p = 0.780$ for EF-Global) (Additional file 1: Fig. S6B).

EF-Focal had a better characteristic change over time toward remission

Symptomatology

Both subtypes showed a significant reduction in PANSS negative, positive, affective, and cognitive symptom scores following treatment ($p_{FDR} < 0.001$) (Fig. 5B; Additional file 1: Tables S13, S14). Significant interactions between subtype and time were observed for the negative ($p_{FDR=}0.025$) and cognitive symptom dimensions ($p_{FDR=}0.036$). Post-hoc analyses revealed that the reduction in negative symptoms over time was significantly more pronounced in EF-Focal than in EF-Global. Specifically, although the two subtypes did not significantly differ in negative symptom dimensional scores at baseline ($p_{FDR=}0.139$), EF-Focal showed significantly lower scores than EF-Global at follow-up ($p_{FDR} < 0.001$). Likewise, no significant difference was detected between subtypes at baseline for cognitive symptoms ($p = 0.193$), but this contrast became significant during follow-up ($p_{FDR} < 0.001$).

Executive functioning

The results showed a significant subtype-by-time interaction ($p_{FDR}=0.003$) for working memory maintenance (Fig. 5B; Tables S13, S14). Post-hoc analyses indicated that, although the two subtypes did not differ significantly in working memory maintenance capacity at baseline ($p_{FDR}=0.246$), EF-Focal had significantly higher scores than EF-Global at follow-up ($p_{FDR} < 0.001$) and its baseline assessment ($p_{FDR=}0.037$). However, no significant improvement was noted for EF-Global patients during follow-up ($p_{FDR=}0.749$).

Peripheral blood indices

We found significant subtype-by-time interactions for the peripheral immune markers lymphocyte count and platelet count, along with the metabolic marker lactate dehydrogenase (all $p_{FDR} < 0.05$). Post-hoc multiple comparisons showed no significant differences between the two subtypes in these markers at baseline ($p_{FDR} > 0.05$) (Fig. 5B and Additional file 1: Table S13). However, at follow-up, EF-Global had significantly higher platelet counts ($p_{FDR} < 0.001$) than EF-Focal and its baseline assessment. Similarly, lactate dehydrogenase levels were significantly higher in EF-Global compared to EF-Focal at follow-up ($p_{FDR} < 0.001$). While there was a non-significant increase in lactate dehydrogenase for EF-Global over time, a reversed pattern was seen in EF-Focal, where the lactate dehydrogenase level at follow-up was significantly decreased from that at baseline ($p_{FDR=}0.003$). The lymphocyte count for patients in EF-Focal was also significantly decreased in re-assessment ($p_{FDR=}0.018$), but, for those in EF-Global, no observation of such a significant reduction was made. Furthermore, we found the change in lactate dehydrogenase levels to significantly mediate the association between EF subtype and the negative symptoms assessed at follow-up (indirect effect: −0.147, 95% CI [−0.331, −0.011], $p = 0.033$) (Fig. 5C).

Collective features of EF subtypes

Overall, patients with EF-Global showed poorer initial conditions and outcomes relative to those with EF-Focal. At baseline, patients with EF-Global showed weaker performance in interference inhibition, working memory updating, and maintenance relative to healthy individuals (Fig. 6A). These patients similarly demonstrated more severe positive and cognitive symptoms, retardation, and anxiety affective symptoms and higher levels of multiple peripheral immune-inflammatory markers (neutrophils, C-reactive protein, NLR, PLR, MLR), and a metabolic index (HDL-C) relative to EF-Focal patients (Fig. 6B). Accordingly, patients within EF-Focal have a better prognosis; they are significantly more likely to achieve remission, and individuals with higher baseline EF-Focal memberships are often classified as in remission (Fig. 6C). EF-Focal patients experience significantly larger reductions in negative and cognitive symptoms, improved working memory maintenance, more stable peripheral immune-inflammatory levels, and superior metabolic functions (lower lactate dehydrogenase levels) over time than those in EF-Global (Fig. 6C).

Supporting findings

1. In *k*-means clustering, the validity indices and stability analysis both pointed to an optimal solution at cluster number 2 (Additional file 1: Fig. S10A-C), which was likewise obtained using fuzzy clustering (Fig. 2). Patients in the two clusters yielding from *k*-means were overlapped with the c-means–derived EF-Focal and EF-Global (78 and 67 patients, respectively). However, as follows: (1) The performance of subtype-based classification was markedly reduced (AUC = 0.61) (Additional file 1: Table S15). (2) The *k*-means method does not provide cluster membership information, and hence it cannot identify cluster-ambiguous samples at baseline. This approach also demonstrated lower longitudinal stability (76/86 [88.37%] patients stable) compared to ambiguous cases filtered out (72/78 [92.31%]) (Additional file 1: Table S15). (3) *K*-means clusters were significantly associated with patients' remission outcomes using two different definition criteria (RSWG: $p = 0.011$; 50% reduction in total PANSS score: $p = 0.029$), albeit less so than in our main results, where this association was significant in three definition criteria (Additional file 1: Fig. S11, Additional file 1: Table S15). (4) *k*-means subtypes failed to capture a significant difference in working

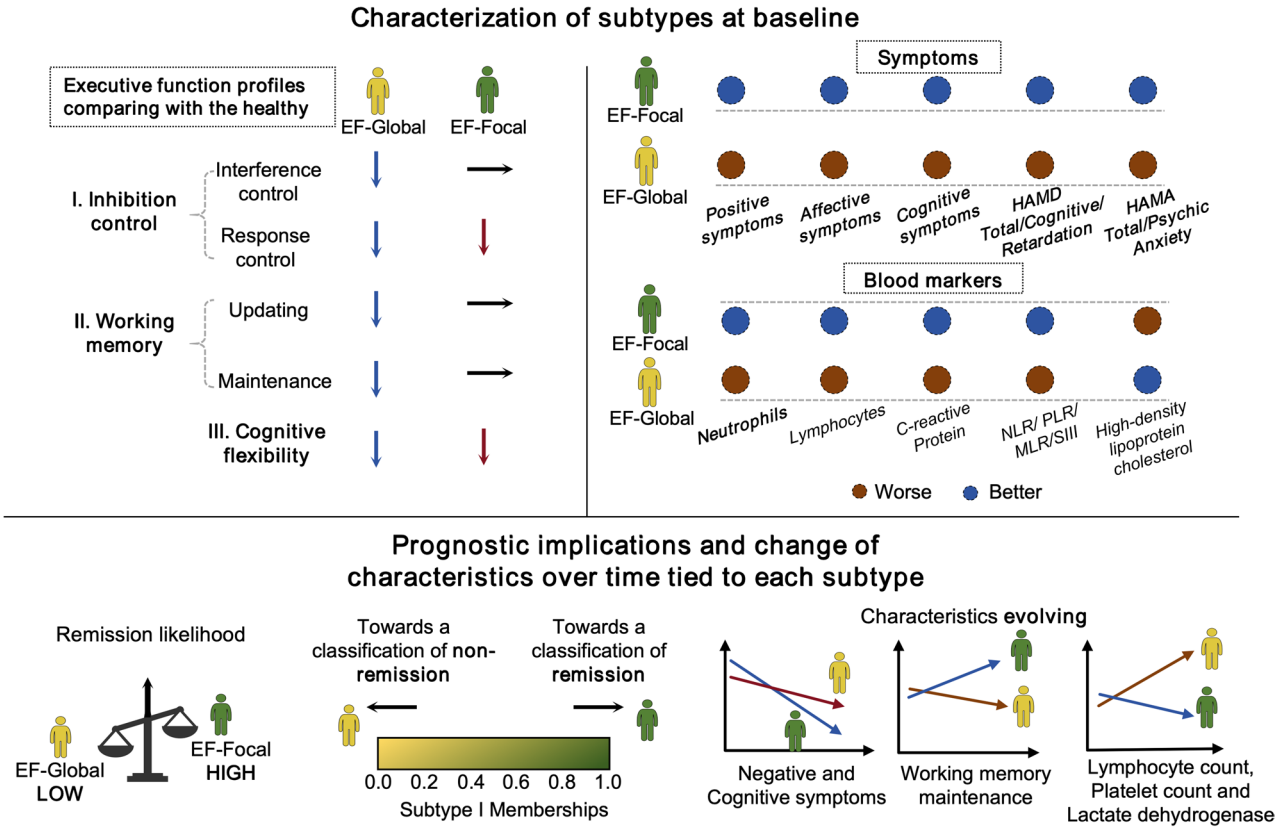


Fig. 6 A summary of subtype characterizations. ***Abbreviations.*** EF, executive function; NLR, Neutrophil-to-lymphocyte ratio; PLR, Platelet-to-lymphocyte ratio; MLR, Monocyte-to-lymphocyte ratio; SII, Systemic immune-inflammation index; HDL-C, High-density lipoprotein cholesterol

memory maintenance capacity between baseline and follow-up assessments (all $p_{FDR} > 0.05$), while the FCM derived EF-Focal but not EF-Global ($p_{FDR=}0.037$), which improved specificity (Additional file 1: Table S15).

2. The patients of core subtypes using different membership thresholds (0.55, 0.65, 0.70) replicated the between-subtype differentiations in baseline clinical symptoms, inflammatory markers, and EFs (except for an absent difference in response inhibition with cutoff 0.65) (Additional file 1: Table S15), as well as remission outcomes (Additional file 1: Fig. S11). For prognostic classification of remission statuses, the main analysis using a membership of < 0.60 as the threshold achieved an AUC of 0.78, which was decreased to 0.71 (0.55 and 0.65 membership cutoffs) and 0.65 (0.7 cutoff) (Additional file 1: Table S15). A similar declining pattern was observed for subtype-based classification with an alternative membership cutoff value (0.7 as the worst: 0.61) (Additional file 1: Table S15).
3. Only one EF variable, which measures interference inhibition, showed a significant correlation with the total PANSS score ($r = 0.3$, $p_{FDR=}0.033$). Adjusting baseline total PANSS score did not alter the significant finding when this score was not controlled in the logistic regression model (all *ps* < 0.033; Additional file 1: Fig. S12). Repeating our clustering approach on EF assessments with additional adjustment for baseline total PANSS score likewise revealed an optimal solution of two clusters (Additional file 1: Fig. S13A), along with an indication of a 0.6 cutoff over the membership values to derive core subtypes (Additional file 1: Fig. S13B). Between-subtype differentiations were also replicated, except for some measurements in EF and blood immune assays, which were significantly (or moderately) correlated with individual baseline total PANSS scores (e.g., RT in incongruent stimuli, NLR, and PLR). The emerged EF subtypes showed comparable performance in subtype-based classification (AUC = 0.78) (Additional file 1: Fig. S13C) but poorer performance in prognostic classification (AUC = 0.70) when compared to our main results (Additional file 1: Fig. S13D).
4. Medication characteristics (type, dosage, concomitant use) did not differ significantly between EF subtypes (all $p_{FDR} > 0.155$; Additional file 1:

Table S2), and none of the 28 tested associations between medication variables and EF performance reached significance (all $p_{FDR} > 0.521$; Additional file 1: Table S11). Subtyping after controlling the effect of antipsychotic medication (OZP dosage) on baseline EF assessments likewise pointed to an optimal solution at K = 2 (best clustering quality achieved with $m = 2.5$ as well) (Additional file 1: Fig. S14). Significant differences in EF domains, symptom profiles and blood markers between the ensuing subtypes replicated our main results (Additional file 1: Fig. S14).

5. The length of the interval between baseline and reassessment does not influence our main results; (i) Individual follow-up duration was not significantly associated with baseline EF (all $ps > 0.23$), re-evaluated EF (all $ps > 0.22$), or the change in EF scores over time (all $ps > 0.14$) (Additional file 1: Fig. S15A). (ii) The time interval between baseline and reassessment did not differ significantly between the two EF subtypes or between remission and non-remission patients (all p-values > 0.28, Wilcoxon rank-sum test) (Additional file 1: Fig. S15B). (iii) Adding the test-retest intervals to the logistic regression model as a covariate-of-no-interest and controlling for it, the significant associations (OR = 0.193, $p = 0.011$) between EF subtype and remission status remained (Additional file 1: Fig. S15C). (iv) Adjusting the time intervals between the test and the retest on FCM membership values first and then establishing classifiers to distinguish between remitted and non-remitted outcomes retained the discriminative power (AUC = 0.73) (Additional file 1: Fig. S15D). (v) The classifier (baseline FCM membership degrees as features to classify remission status as in the original manuscript) trained on the shorter-follow-up cohort successfully classified remission outcomes in the longer-follow-up cohort, achieving an AUC of 0.72 (Additional file 1: Fig. S15E).

Discussion

In this study, we used two fuzzy clustering methods to derive core subtypes for schizophrenia based on EF dimensions assessed using five well-established behavioral paradigms. The findings' replicability was confirmed with an independent "recurring local validation" sample. Our results pointed to a differentiation between the EF-Focal and EF-Global subtypes in working memory updating function, along with multiple symptoms and inflammatory and metabolic indices. Using individual EF features from the main sample moreover allowed an accurate classification of patient subtypes in the independent sample. Importantly, the subtypes showed varying responses to treatment, with EF-Focal more likely to achieve remission, and this difference was not attributed to a difference in overall severity at baseline. Using baseline FCM membership values, remission statuses can be classified with a high discriminative power. Finally, we observed remitted patients to experience significantly larger reductions in negative and cognitive symptoms, improved working memory maintenance, more stable peripheral inflammatory levels, and superior metabolic functions over time.

The importance of cognitive impairments in schizophrenia and their heterogeneity in severity across individual patients have promoted attempts to derive subgroups based on neuropsychological assessments. These attempts have used comprehensive cognitive batteries covering multiple domains that are thought to be affected in patients with schizophrenia. Using cluster analysis and latent profile analysis, previous studies have yielded inconsistent numbers (2–4) of cognitive subgroups according to the varying assessment tools adopted and patient cohorts recruited. These cognitive subgroups generally include a virtually intact and a globally impaired cohort, respectively, with also an intermediate (or selective) impairment cluster usually produced [23, 27, 31]. Interestingly, such an intermediate subgroup has been further divided by a differential performance in EFs [23, 27, 30], pointing to a critical role of EF in dissecting the cognitive heterogeneity in schizophrenia. However, previous studies have treated EF as a unidimensional construct when leveraging a broad range of cognitive assessments that overlooked the multi-dimensional aspects of EF. Here, we tapped into the neuropsychological processes underlying three EF dimensions with seven variables derived from five behavioral paradigms and used a fuzzy-clustering approach to depict the heterogeneous EF profile in schizophrenia. In particular, existing literature consistently highlights that inhibition function, cognitive flexibility, and working memory exhibit significant inter-individual heterogeneity, with patient performance ranging from preserved to severely impaired [23, 34, 106]. Variations in EF subdomains have been shown to have genetic [22] and neurobiological underpinnings [128], contributing to different psychopathological processes [39]. Indeed, patients with a severe deficit in working memory are found to have high positive symptoms and low cognitive functioning [129]. Our EF subtypes differed in baseline positive and cognitive symptoms, along with affective problems. Also, there are prior observations that negative and disorganized symptoms are associated with distinctive working memory deficits in patients with schizophrenia [129]. Our longitudinal analyses extended to a dynamic association between working memory maintenance (relative to updating, as more evidently differed between subtypes at baseline)

performance improvement and the reduction of negative and cognitive symptoms in one of the subtypes (with a tendency toward remission) over time. Previous research has linked better executive functioning to a more likely remission outcome in schizophrenia [130, 131]. In particular, specific executive processes, such as interference inhibition and working memory, have been strongly associated with symptom remission [46, 48, 132]. Delving into the mechanisms underlying such relationships pointed to multiple processes that may contribute to the emergence and aggregation of several behavioral symptoms. For example, impaired working memory could undermine patients' capacity to maintain and manipulate mental representations, affecting abstract thoughts and reasoning as well as the processing of emotional cues, thus increasing the risk of developing negative symptoms (e.g., apathy and diminished expressive behavior) [133, 134] while aggravating cognitive and affective symptoms [129, 135].

Moreover, EF-Global patients exhibited significantly elevated baseline inflammatory levels (e.g., higher NLR and C-reactive protein levels) and reduced metabolic function (e.g., lower high-density lipoprotein levels) than those patients with EF-Focal. Elevated inflammatory levels have been associated with impairments in specific cognitive functions, such as inhibitory function [55, 136, 137] and working memory [138]. Moreover, patient subgroups of schizophrenia with varying levels of inflammation were also found to be distinctive in working memory. This relationship is likely mediated by inflammation's effects on the function of the central executive network in the brain. This effect from peripheral inflammation can occur through multiple pathways, as detailed in previous studies on brain–immune cross-talk [136]. Neuroinflammation is also common in schizophrenia, which may exert a more direct impact on functional brain dysconnectivity, thus reducing cognitive functions [139]. Furthermore, antipsychotics can worsen lipid metabolism while lowering high-density lipoprotein, which has been related to treatment responses, e.g., symptom reduction and daily function recovery.

Importantly, patients in EF-Focal, having higher rates of remission, experienced higher lymphocyte counts than those in EF-Global, which means that individuals in EF-Global had higher levels of immune abnormalities. Previous studies demonstrate that lymphocyte counts are associated with immune dysregulation in schizophrenia [140]. Also, higher lymphocyte counts, along with a better NLR, are related to overall symptom severity as well as negative and depressive symptoms [52, 141, 142]. Our results suggest that a lower baseline NLR level and reductions in lymphocyte counts over time relate to more alleviated negative and cognitive symptoms in patients of EF-Focal. Furthermore, specifically the level of the metabolic index lactate dehydrogenase is found to significantly mediate the association between EF subtypes and their negative symptoms at follow-up. However, the effect sizes of mediation observed in this analysis were small (the indirect effect accounted for 6% of the total effect), possibly due to the fact that many factors beyond blood markers can mediate such a relationship between subtypes and symptom reductions over time. Previous studies with single inflammatory and metabolic indices as the mediator have likewise revealed low effect sizes of < 9% in schizophrenia [143, 144]. A decrease in lactate dehydrogenase was evident in the milder EF-Focal population within schizophrenia. Conversely, an increase in lactate dehydrogenase (e.g., in our EF-Global patients) has been shown to indicate adverse outcomes, such as prominent extra-pyramidal symptoms and the occurrence of violence [145, 146], highlighting the importance to regulate this bio-factor. Specifically, previous research advises switching to antipsychotics with a milder impact on the mitochondrial respiratory chain if patients exhibit elevated lactate levels shortly after initiating treatment. Furthermore, Fang et al. (2025) suggest that pharmacologically inhibiting lactate production with agents such as dichloroacetate could reprogram cellular metabolism [147]; this approach may alleviate the behavioral and neuronal deficits associated with treatment-resistant schizophrenia.

Methodologically, our use of a fuzzy clustering approach to identify cluster-ambiguous patients along continuous EF dimensions matches with a heterogeneous construct of schizophrenia psychopathology and resonates well with the notion that some patients with schizophrenia may not have a clear subtype attribution but lie between more differentiated subtypes or deviate from them given a possibly high continuum in this disorder [148]. In contrast, previous subtyping investigations relied on hard clustering methods (e.g., *k*-means clustering), which assume that each patient has a cluster attribute and the constituted subtypes represent separate, non-overlapping groups [27, 31]. While clustering provides common patterns, binary divisions would contradict the continuous individual expression along EF dimensions given that EF impairments in patients with schizophrenia represent a spectrum. The heterogeneity is reflected in a varying degree of such impairments across dimensions. Therefore, a soft-clustering approach that allows each individual to belong to multiple clusters with varying membership degrees better aligns with the differential individual expression level in each EF dimension. In particular, this allowed for filtering out cluster-ambiguous patients to derive "core" subtypes and warrants a high longitudinal stability as those intermediate cases could be more changeable in their cognitive profiles over time. This is true in both our

results using *k*-means clustering and those reported in previous studies that have established cognitive subtypes (e.g., BACS and MATRICS-based clusters) using hard-assignment approaches (~87% or 80% maintenance in a 4- or 12-week interval) [23, 27]. However, most of the previous studies did not incorporate a follow-up session. Importantly, this precluded an evaluation of how cognitive subtypes might predict prognostic statuses. Moreover, our application of *k*-means clustering for a direct comparison showed the ensuing subtypes more difficult to be classified from baseline EF assessments, and less robustness to predict outcomes using different ways of defining remission and a lower specificity in capturing a differential change of performance in EF over time. The superiorities of using fuzzy clustering we showed here resonate well with previous practices in schizophrenia [63] and beyond [149, 150], i.e., that this approach derived clusters with enhanced biological interpretability and clinical usefulness to enable a more accurate representation of individuals' disease trajectories where subjects can have mixed characteristics [151]. Furthermore, the membership values yielded from baseline data using fuzzy clustering allowed a discrimination of remission statuses when re-assessed. Interestingly, our subtype-based diagnostic classification, which treated EF-Global as the positive class and EF-Focal as the negative class, yielded greater sensitivity (0.79) for detecting positive cases with a poorer remission outcome (Fig. 3D and Additional file 1: Table S12). This indicated that our machine learning classifier is particularly adept at detecting patients who are more severe at baseline and are more likely not to be remitted as it is able to classify them as negatives. Clinicians may leverage this model to identify patients who are more likely to experience poor treatment response at earlier stages using the current model. Specifically, our EF-based subtyping would address a critical gap in current practice where psychotic disorders are typically managed uniformly and personalized care is often a product of "trial and error" [152]. This "fail-first" approach necessitates lengthy evaluations of standard treatments before alternatives are considered [153], delaying the optimal treatment window. Our framework, in turn, enables prognostic stratification upon initial presentation, empowering clinicians to implement preventive interventions for those most likely to experience poor outcomes. Importantly, our behavioral paradigm is streamlined into a 20–30-minute assessment, making it feasible for standard clinical settings. In particular, for patients assigned to the EF-Global subtype (associated with a poor prognosis), more intensified strategies such as antipsychotic polypharmacy—which has been associated with reduced risk of re-hospitalization [154, 155]—could be prioritized, along with transcranial magnetic stimulation, which serves as an effective augmentation strategy for partial responders [156]. Furthermore, we observed an improvement in working memory in the EF-Focal subtype to track with a symptomatic remission over time. This suggests that cognitive enhancement facilitates recovery; therefore, cognitive remediation therapy and pro-cognitive pharmacological agents (e.g., adjunctive memantine) [157, 158] are also recommended to improve symptomatic remission rates in schizophrenia [73, 159].

Several limitations can be mentioned. First, although creating a subtyping framework that is generalizable across various clinical sites and diverse patient populations would be ideal and remarkable, we here adopted a "recurring local validation" method, which involved an independent sample collected at different points within the same medical center. Importantly, this approach has gained increased appreciation in recent years due to repeated practical challenges in developing generic models applicable in clinical psychiatry. Notably, it is suggested to be a means to improve the utility and feasibility of developing machine learning tools that support local medical services. Our strategy of "recurring local validation" hence aligns with emerging artificial intelligence–driven healthcare paradigms, which aim to prevent performance variations and (most likely) degradation across different populations and facilities [160]. Nevertheless, future studies incorporating data from other ethnicities and healthcare systems are required to verify the generalizability of the currently developed EF subtypes. To this end, future studies might consider incorporating additional sites in their analyses. Methods such as federated learning can help fine-tune and optimize machine learning models to enhance their performance and generalizability without the need to share individual raw data, thus addressing potential ethical concerns and data privacy constraints across clinical centers [161]. Second, we used detailed EF assessments for subtyping schizophrenia, in contrast with previous studies that evaluated EF as a single construct. Future research could incorporate behavioral assessments from other cognitive domains to explore whether additional subtypes might emerge, allowing for finer differentiation within the heterogeneous patient population. Third, the present work examined several inflammatory and metabolic indices to explore their relations with EF subtypes and corresponding prognostic statuses, but future research is encouraged to evaluate additional factors like glucose and amino acid metabolism (e.g., glutamate) [162] and pro-inflammatory cytokines (e.g., interleukin-6, interleukin-1β, and tumor necrosis factor–α), which have been associated with schizophrenia's pathophysiological underpinnings, treatment responses, and psychopathological processes [138]. Therefore, these evaluations will help delineate a more comprehensive immunometabolic profile to better characterize cognitive-based schizophrenia subtypes. Fourth,

while there is evidence showing that the performance of outcome prediction remains comparable between short-term 4-week and long-term 52-week follow-up endpoints [69], whether our subtypes differing in EF deficient profiles would indicate long-term prognostic statuses remains for future studies with a longer follow-up duration and more reassessment timepoints to investigate. Interestingly, several previous studies showed that short-term improvement in symptoms can predict subsequent outcome statuses, highlighting the value of monitoring response earlier in the course of treatment [163–165]. Additionally, our classifiers trained on shorter-follow-up patients (< 35 days) allowed for discriminating the remission statuses of those with longer follow-up times (> 35 days) (Additional file 1: Fig. S15E), probably holding promise for application in patient cohorts with longer follow-up periods. This aligns with previous findings in schizophrenia research, where models trained on short-term responses (2–4 weeks) successfully predicted long-term outcomes (up to 52 weeks) [153, 166]. Fifth, while different lines of evidence highlight that cognitive function shares a fundamental—and potentially causal—link with the disease process, the current analyses established association patterns rather than direct causality [167]. Future studies incorporating controlled medication regimens, longer follow-up periods, and more frequent reassessment points are encouraged to elucidate the potential causal relationships between cognitive subtypes and symptom remission in schizophrenia. Finally, although our attrition rate aligns with previous reports [168, 169], it raises questions about potential selection bias. However, our complementary analyses demonstrated that the followed patients represented a more general patient cohort with respect to baseline characteristics and the distribution of FCM membership degrees, including those who were lost to follow-up. Moreover, the prognostic classifier, which was trained on reassessed patients, showed similar proportions of discriminated remitted versus non-remitted conditions in participants lost to follow-up compared to those observed in the reassessed patients (Additional file 1: Fig. S6B). Nevertheless, future studies should focus more on managing attrition rates, even though increasing patient adherence to longitudinal study designs remains challenging.

Conclusions

In conclusion, multi-dimensional EF assessments differentiated two longitudinally stable subtypes for schizophrenia. These subtypes captured differences in baseline symptomatic and peripheral immunometabolic features, as well as (importantly) prognostic implications, with one subtype associated with greater symptom reduction, EF performance improvement, and metabolic function recovery over time. Baseline EF assessments moreover facilitated out-of-sample subtype classification, where the individual membership degrees from our fuzzy clustering approach successfully differentiated patients' remission statuses. Together, these results may assist in stratifying the schizophrenia population for targeted treatments and offer insights into identifying patients more likely to have poor prognoses, allowing for the application of effective interventions at an earlier stage.

Abbreviations

AUC	Area under the receiver operating characteristic curve
BAC	Balanced accuracy
CI	Confidence interval
DCTS	Dimensions and Clustering Tool for Schizophrenia Symptomatology
EF	Executive function
FCM	Fuzzy c-means
GMM	Gaussian mixture model
HAMA	Hamilton Anxiety Rating Scale
HAMD	Hamilton Depression Rating Scale
HC	Healthy control
HDL-C	High-density lipoprotein cholesterol
MLR	Monocyte-to-lymphocyte ratio
NLR	Neutrophil-to-lymphocyte ratio
OZP	Olanzapine
PANSS	Positive and Negative Syndrome Scale
PLR	Platelet-to-lymphocyte ratio
RSWG	Remission in Schizophrenia Working Group
RT	Reaction time
SD	Standard deviation
SE	Standard error

Supplementary Information

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Supplementary Material 1: Additional file 1: "Supplementary Methods", "Figures S1–S15", and "Tables S1–S17.

Supplementary Material 2: Additional file 2: the STROBE checklist.

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Author contributions

T.Y.Z., J.C. and X.Z. conceived and designed the study. T.Y.Z. and X.N.H. acquired and curated the clinical, neuropsychological, and inflammatory data. T.Y.Z. performed the clustering and longitudinal validation analyses. J.C. and X.Z. supervised the methodological development and statistical interpretation. T.Y.Z., J.C. and X.Z. drafted the manuscript. All authors critically revised the manuscript for important intellectual content and approved the final version to be published. All authors read and approved the final manuscript.

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Data availability

Information for the samples used in the present study have been included in Additional file 1. The raw data of these samples are protected and are not publicly available due to data privacy. These data can be accessed upon reasonable request to the corresponding author (X. Z., psyzhaoxin@nwnu.edu.cn). Derived data supporting the findings of this study are available from the corresponding authors (X. Z., psyzhaoxin@nwnu.edu.cn) upon request. The main sample and the followed-up data have been included in a work we previously published [170].

Code Availability
Scripts to run the main analyses have been made publicly available and can be accessed at https://github.com/tyzhang98/Code_Heterogeneous_EFs_in_SCZ.

Declarations

Ethics approval and consent to participate
This study was conducted in full compliance with the appropriate ethical guidelines and was approved by the Ethics Committees of the Third People's Hospital of Lanzhou City and Northwest Normal University (Lanzhou, China; ethics approval nos. TPHL202303 and NWNU202404). All participants provided written informed consent after receiving comprehensive information about the study aims, procedures, potential risks, and benefits. The study adhered to the principles outlined in the Declaration of Helsinki.

Consent for publication
Not applicable.

Conflict of interest
We declare that Dr. Ji Chen is an Editorial Board Member of *BMC Medicine*. The authors declare no competing interests.

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